

BITWISE OPERATORS

Bitwise operator's works on bits.

Turbo-c is a 16 bit compiler. Due to this bitwise operations are limited to 16 bits only [2^0 to 2^{15}].

Bitwise operators operate **integer** type values only.

We have to calculate only the **on** bits [**1**].

When the first bit[**Sign bit**] is **1** then the number is **Negative** and it is **0** then the number is **positive**.

They are very much used in system software development.

Note: Bitwise operator is low level feature.

C-Language supports following bitwise operators.

& -Bitwise and

| - Bitwise or

^ - XOR ==> Exclusive OR

~ - Compliment operator

<< - Left shift operator

>> - Right shift operator

& - Bitwise and: In this both bits are 1's then result bit is 1. Otherwise result bit is 0.

Eg: **25 & 15 = 9**

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111

$$\begin{array}{r} 2 \overline{) 25} \\ 2 \overline{) 12 - 1} \\ 2 \overline{) 6 - 0} \\ 2 \overline{) 3 - 0} \\ 1 - 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 15} \\ 2 \overline{) 7 - 1} \\ 2 \overline{) 3 - 1} \\ 1 - 1 \end{array}$$

25 & 15 = 9

25 = 0000 0000 0001 1001
15 = 0000 0000 0000 1111

&

$$\begin{array}{r} \hline 0000 \ 0000 \ 0000 \ 1001 \\ \downarrow \quad \downarrow \\ 2^3 + 2^0 \\ \downarrow \quad \downarrow \\ 8 + 1 = 9 \end{array}$$

| - Bitwise or: In this both bits are 0's then result bit is 0. Otherwise result bit is 1.

Eg: $25 \mid 15 = 31$

$$25 \mid 15 = 31$$

25 = 0000 0000 0001 1001

15 = 0000 0000 0000 1111

0000 0000 0001 1111

$2^4 + 2^3 + 2^2 + 2^1 + 2^0$

$16 + 8 + 4 + 2 + 1 = 31$

^ - XOR [Exclusive OR]: In this both bits are same then result bit is 0. Otherwise result bit is 1.

Eg: $25 \wedge 15 = 22$

$$25 \wedge 15 = 22$$

25 = 0000 0000 0001 1001

15 = 0000 0000 0000 1111

$\wedge$

0000 0000 0001 0110

$$\begin{array}{r}
 2^4 + 2^2 + 2^1 \\
 16 + 4 + 2 = 22
 \end{array}$$

~ - Complement operator: In complement operation the bits are complimented. i.e. 1's become 0's and 0's become 1's. Due to this +Ve no becomes -Ve and -Ve no becomes +Ve.

eg: ~25 → -26

25 = 0000 0000 0001 1001
1111 1111 1110 0110
-128+64+32+4+2=-26
-128 + 102 = -26

$$25 = 0000\ 0000\ 0001\ 1001$$

$$\sim = 1111\ 1111\ 1110\ 0110$$

$$\begin{array}{c} \backslash \quad \backslash \\ 5 \quad 2 \end{array}$$

$$2+4+32+64+128+256+512+1024+2048+4096+8192+16384-32768=-26$$

$$\sim -25 = 0000\ 0000\ 0001\ 1001$$

$$1's\ \sim = 1111\ 1111\ 1110\ 0110$$

$$2's\ \sim = \begin{array}{r} 0000\ 0000\ 0000\ 0001 \\ 1111\ 1111\ 1110\ 0111 \end{array}$$

$$\begin{array}{c} \swarrow \quad \searrow \\ 24 \quad 2 \\ 16+8=24 \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 01 \\ 0 & 1 & 0 & 1 \\ \hline 1 & 1 & 0 & 10 \end{array}$$

Note: When starting bit is 1 given no is –Ve.

Eg: $\sim -25 \rightarrow +24$

$$\sim -25 = +24$$

$$25 = \begin{array}{|c|c|} \hline 0000 & 0000 \\ \hline \end{array} \begin{array}{c} 0001\ 1001 \\ 1111\ 1111 \end{array}$$

$$\begin{array}{c} 1111\ 1111\ 1110\ 0110 \\ +1 \end{array}$$

$\Leftarrow$ 1's complement

$\Leftarrow$ 2's Complement

$$\begin{array}{c} \sim \\ \rightarrow \end{array} \begin{array}{c} 1111\ 1111\ 1110\ 0111 \\ 0000\ 0000\ 0001\ 1000 \end{array}$$

$$\begin{array}{c} \downarrow \quad \downarrow \end{array}$$

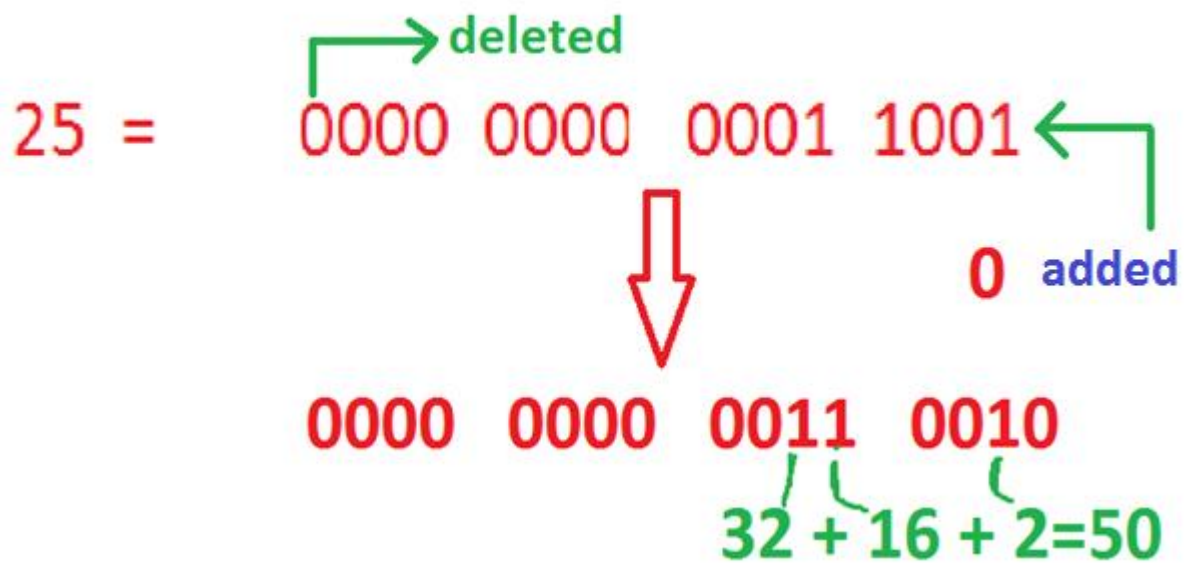
$$16+8=24$$

<< - left shift operator:

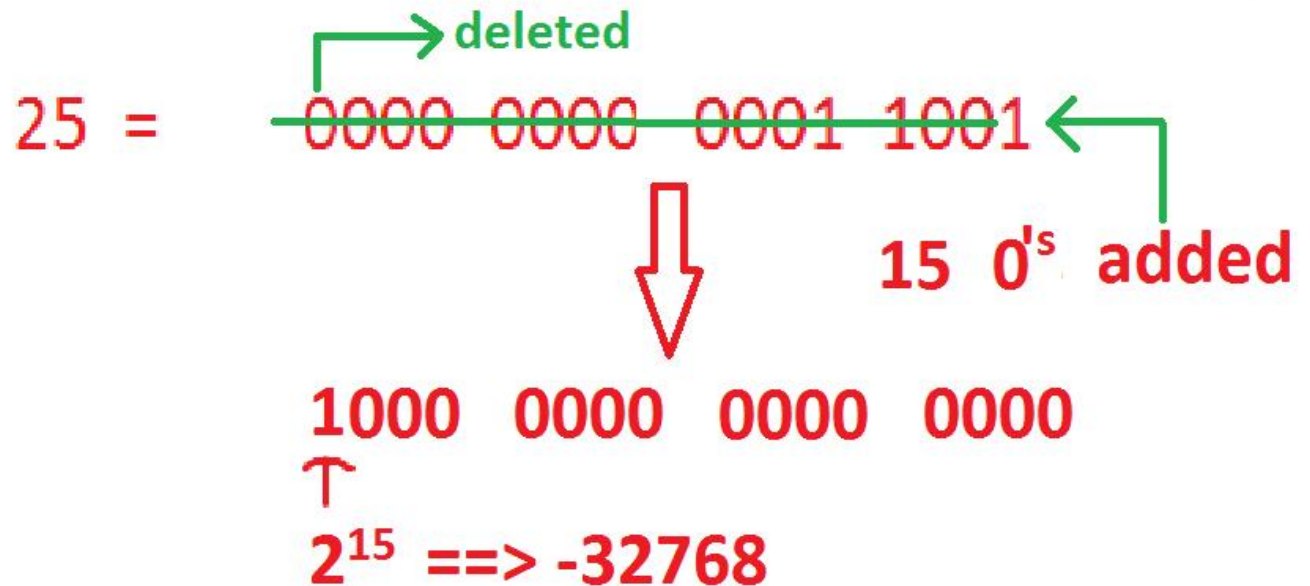
In left shift operation, the specified no of bits are deleted from left side and the same no of **zeros** added on right side. In left shift operation, most probably the value is multiplied with 2 that no of times.

Eg: $25 \ll 1 = 50$, $25 \ll 2 = 100$, $25 \ll 15 = -32768$, $25 \ll 16 = 0$

eg: $25 \ll 1 = 50$



eg: $25 \ll 15 = -32768$



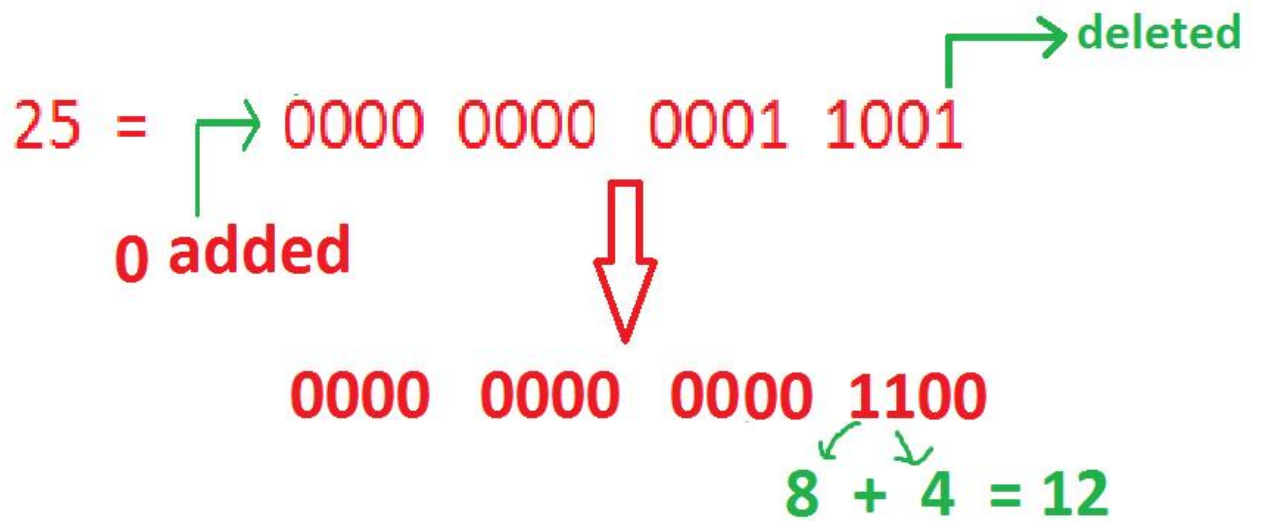
Note: When starting bit 1 no is negative.

>> - Right shift operator:

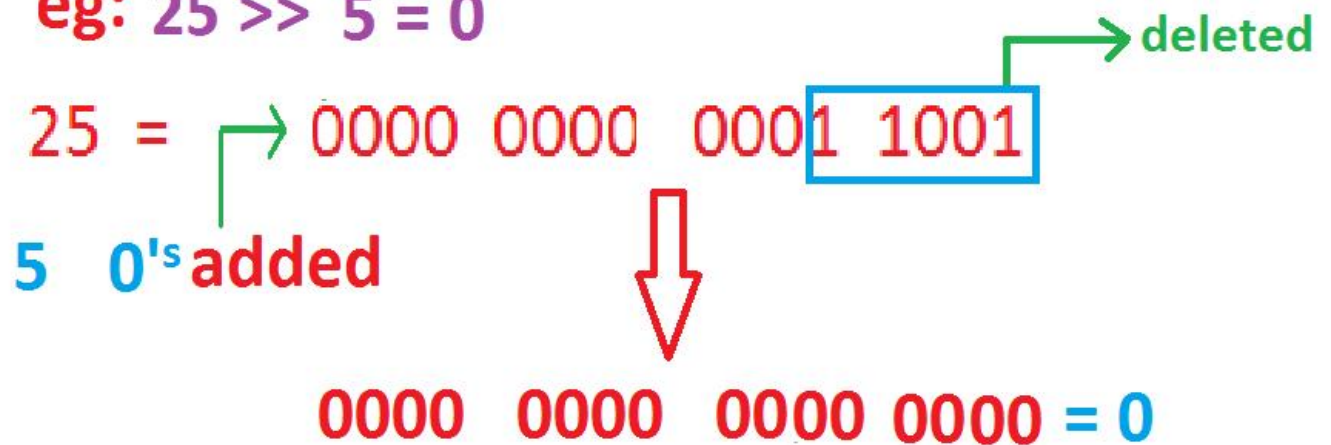
In right shift operation, the bits are moved to right side i.e. the specified no. of bits are deleted from right side and same no. of **zero's** are added left side. Due to this always the number is divided with 2 that no of times.

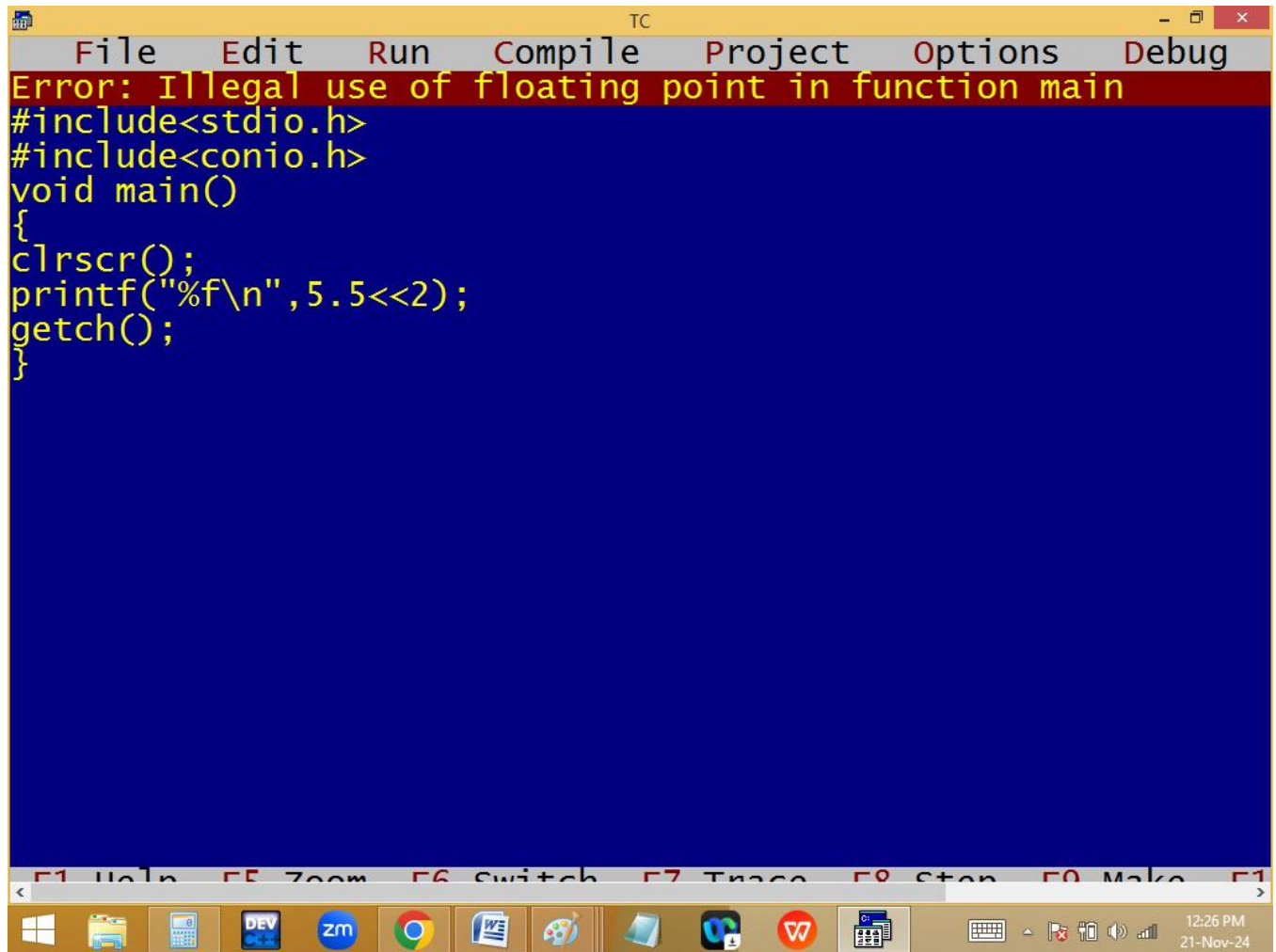
Eg: $25 \gg 1 = 12$, $25 \gg 2 = 6$, $25 \gg 3 = 3$, $25 \gg 4 = 1$, $25 \gg 5 = 0$

eg: $25 \gg 1 = 12$



eg: $25 \gg 5 = 0$

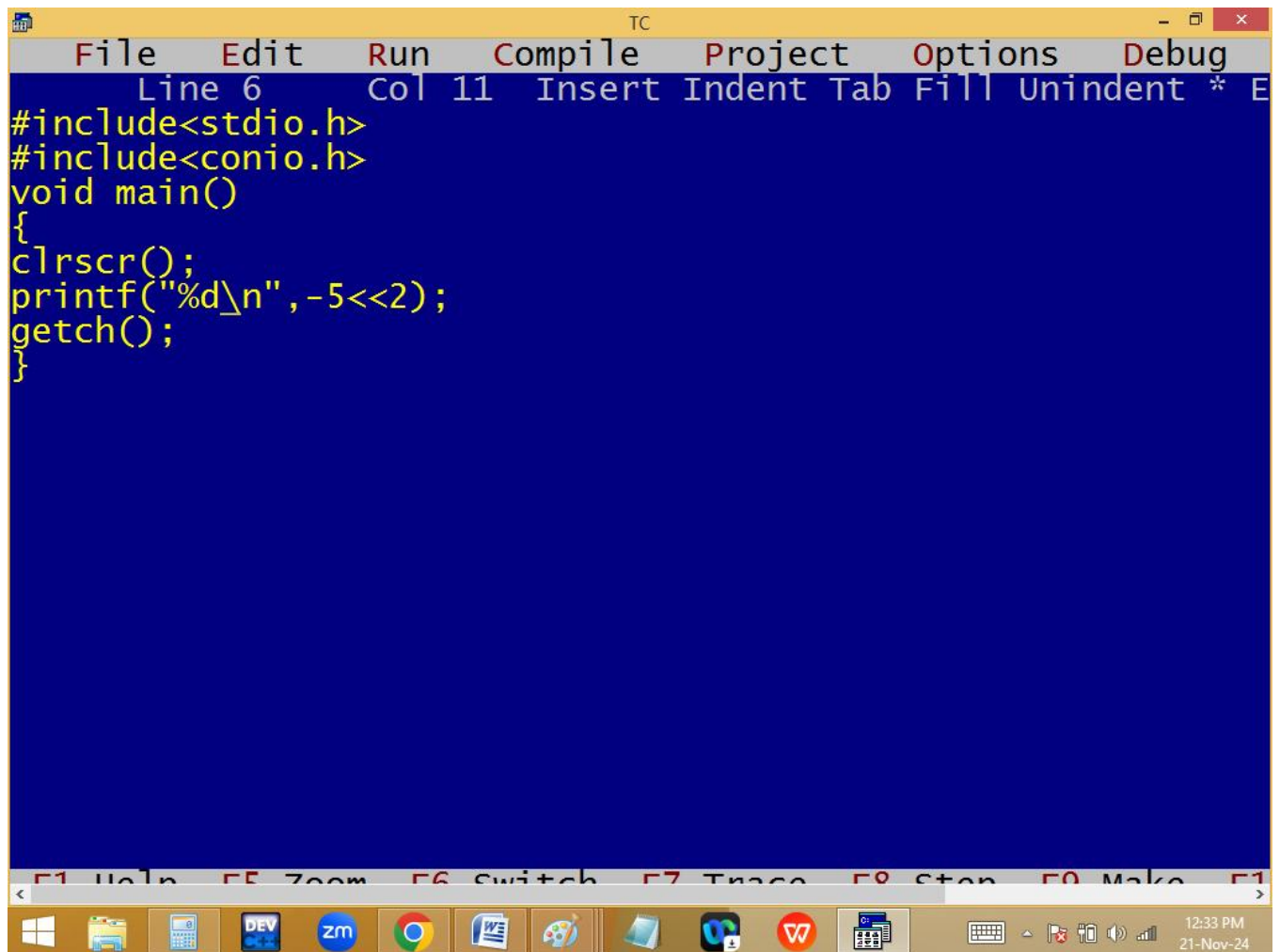




The image shows a screenshot of the Turbo C++ (TC) IDE. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error message banner at the top of the code editor states: "Error: Illegal use of floating point in function main". The code editor has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%f\n",5.5<<2);
getch();
}
```

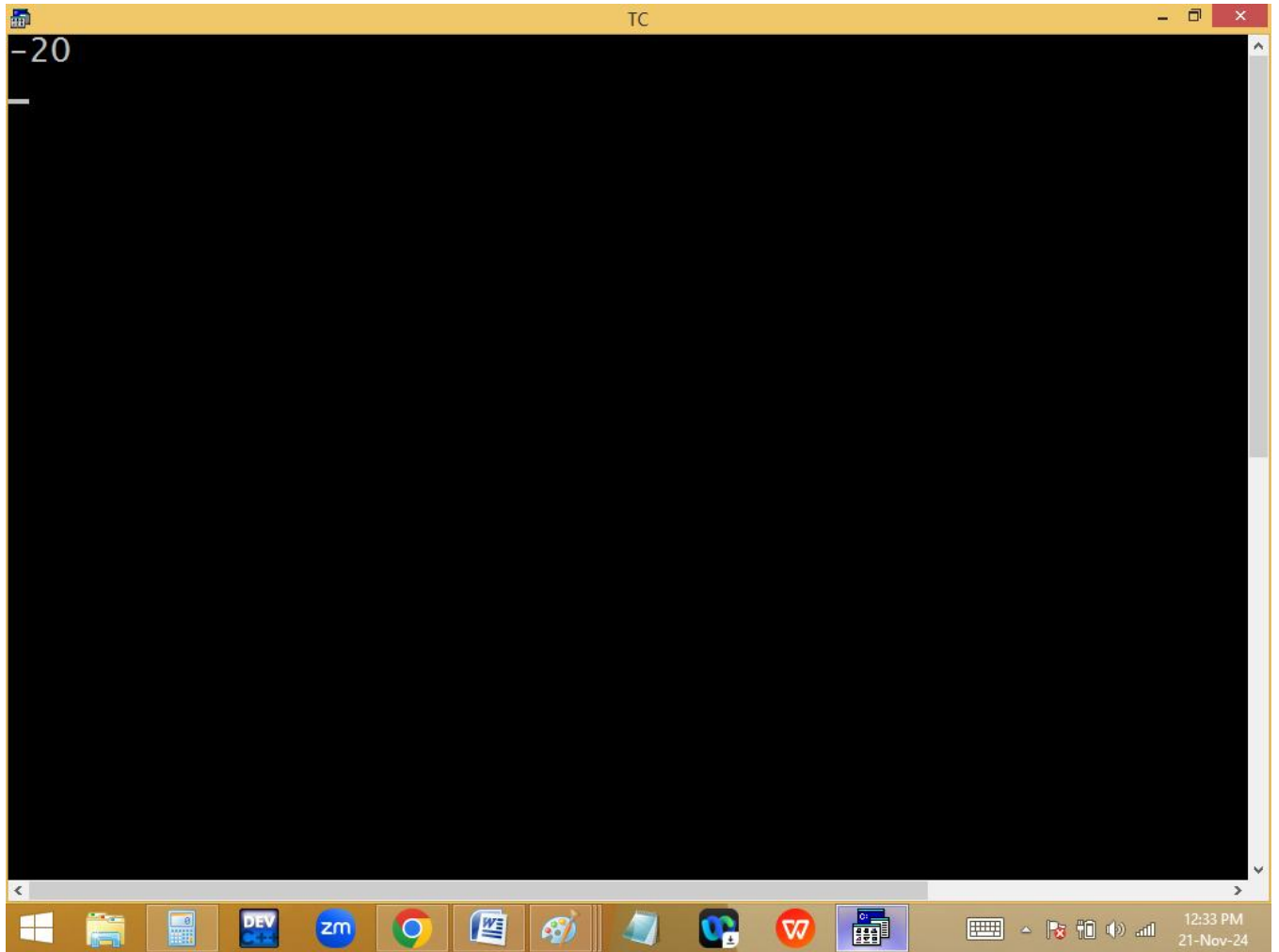
Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, Task Manager, DEV C++, Zoom, Google Chrome, Word, and other applications. The system clock in the bottom right corner indicates the time is 12:26 PM on 21-Nov-24.

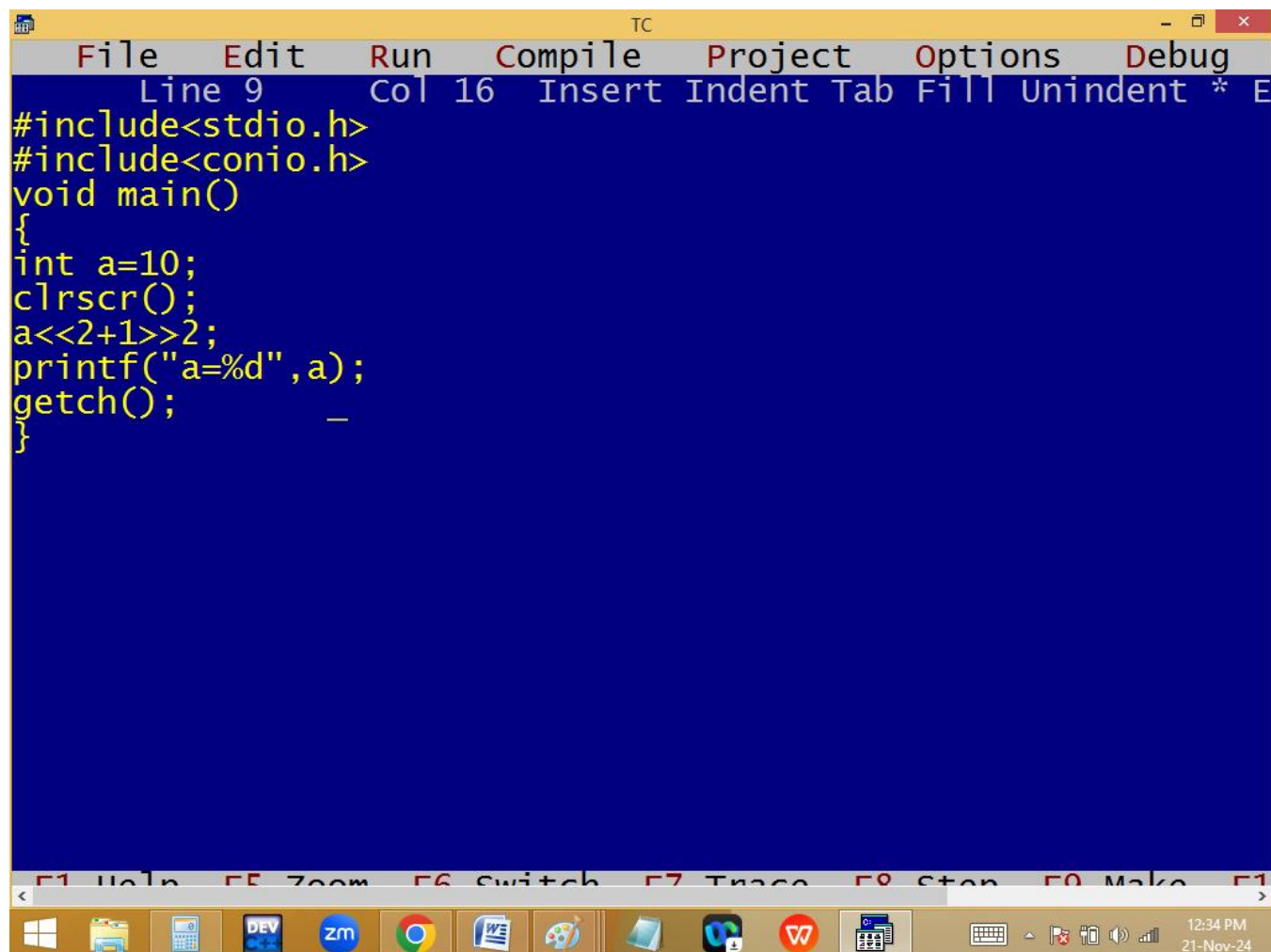


```
TC
File Edit Run Compile Project Options Debug
Line 6 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n",-5<<2);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:33 PM
21-Nov-24

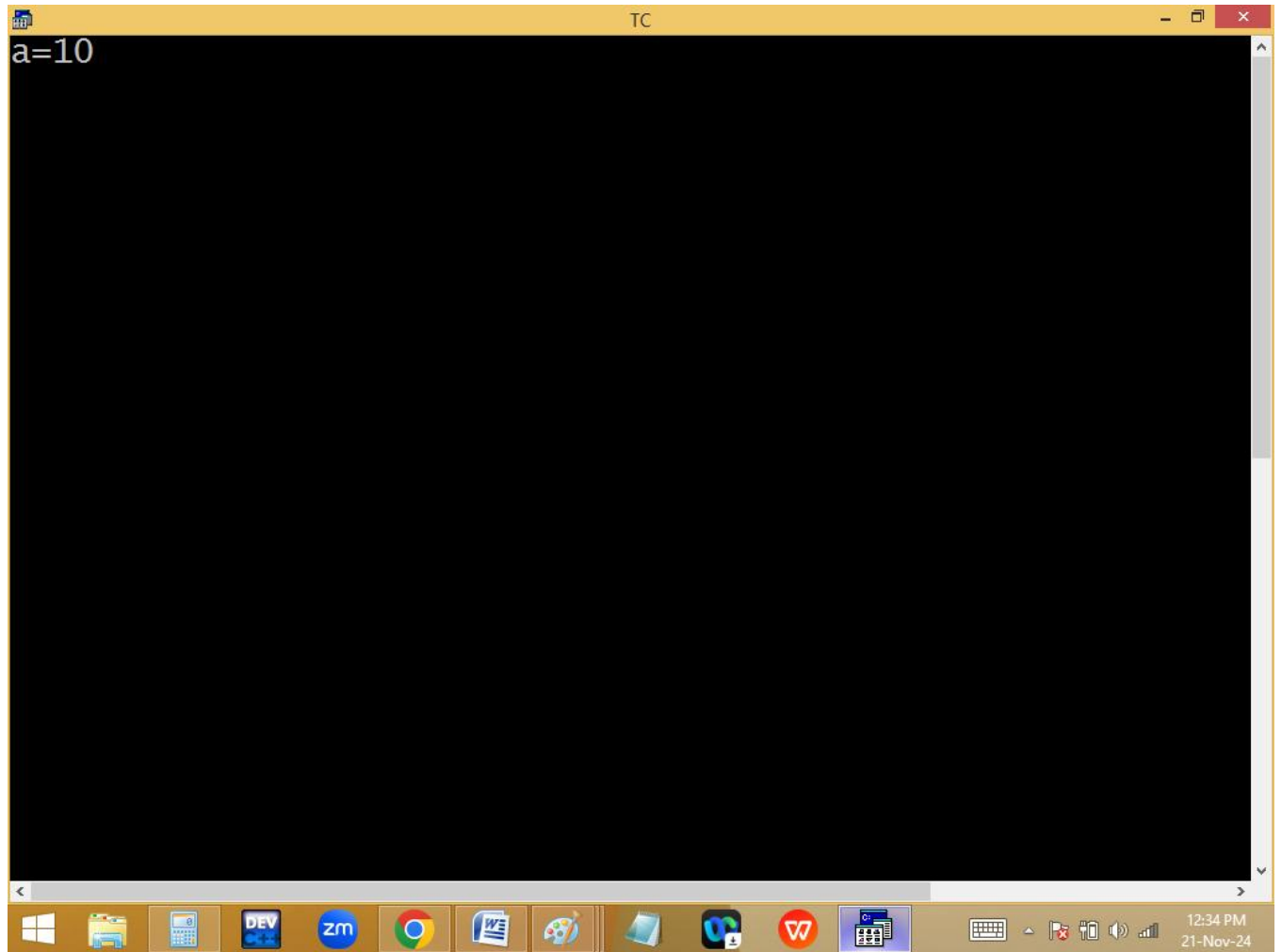


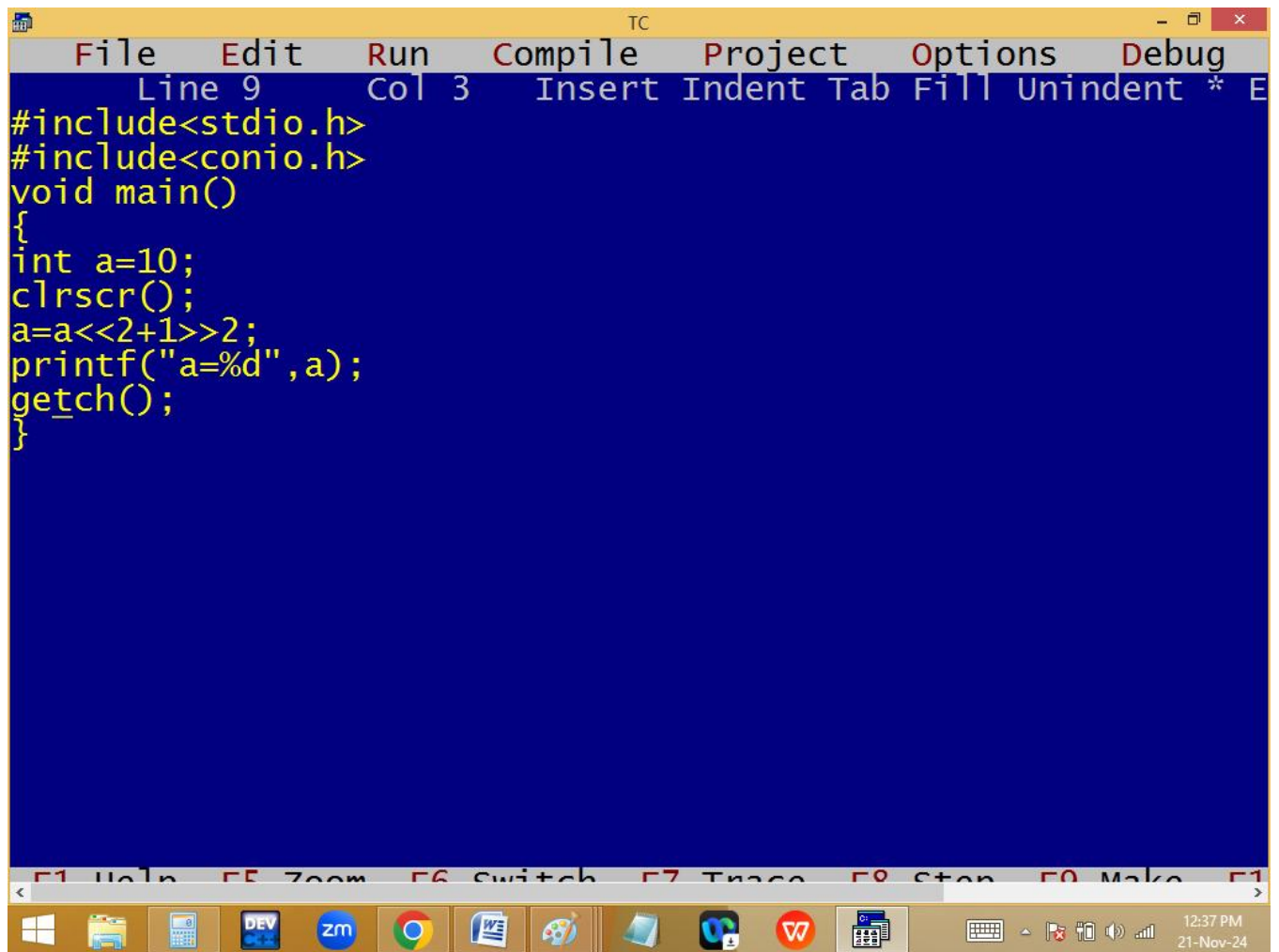


```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 16 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a<<2+1>>2;
printf("a=%d",a);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:34 PM
21-Nov-24

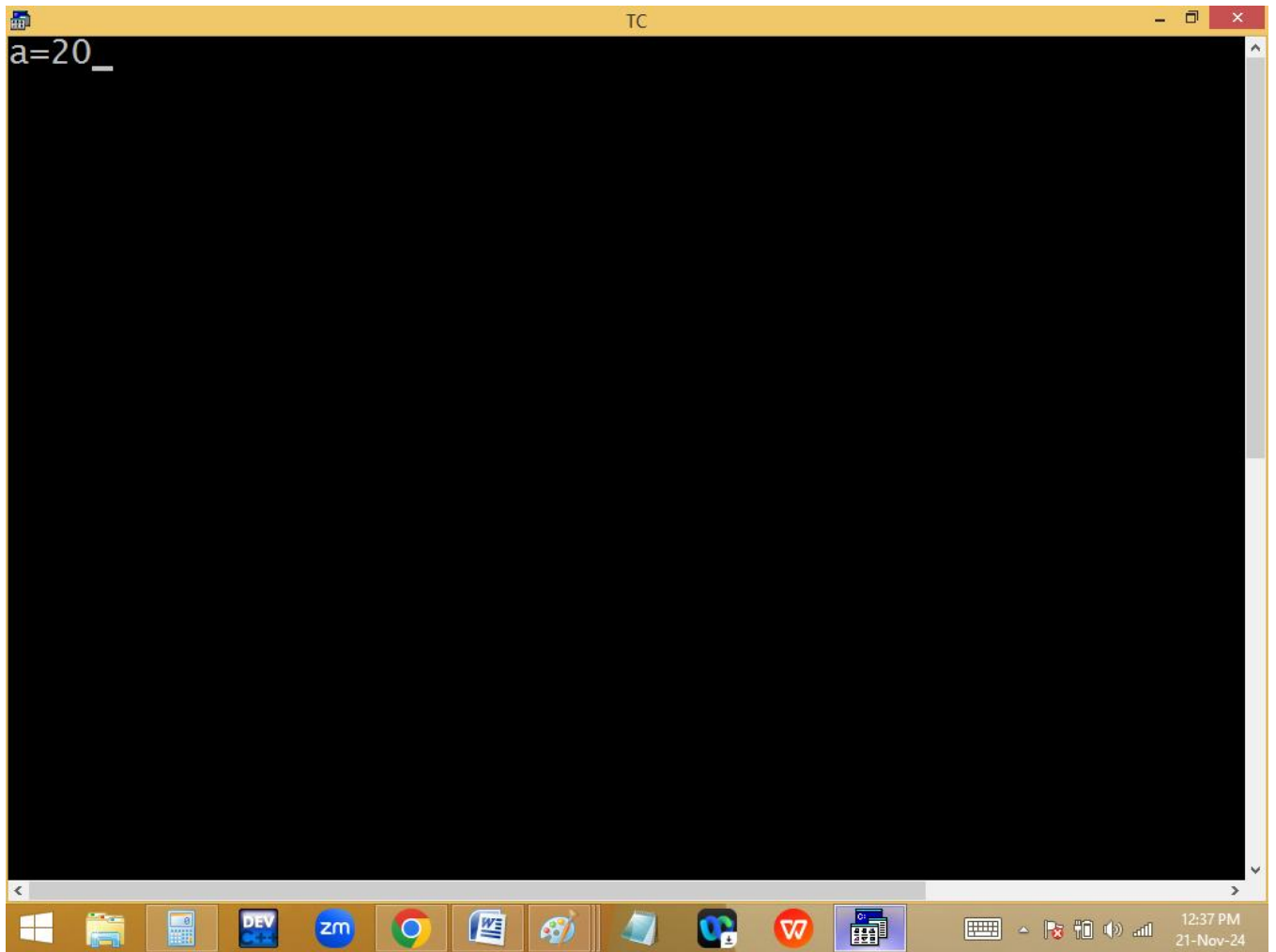


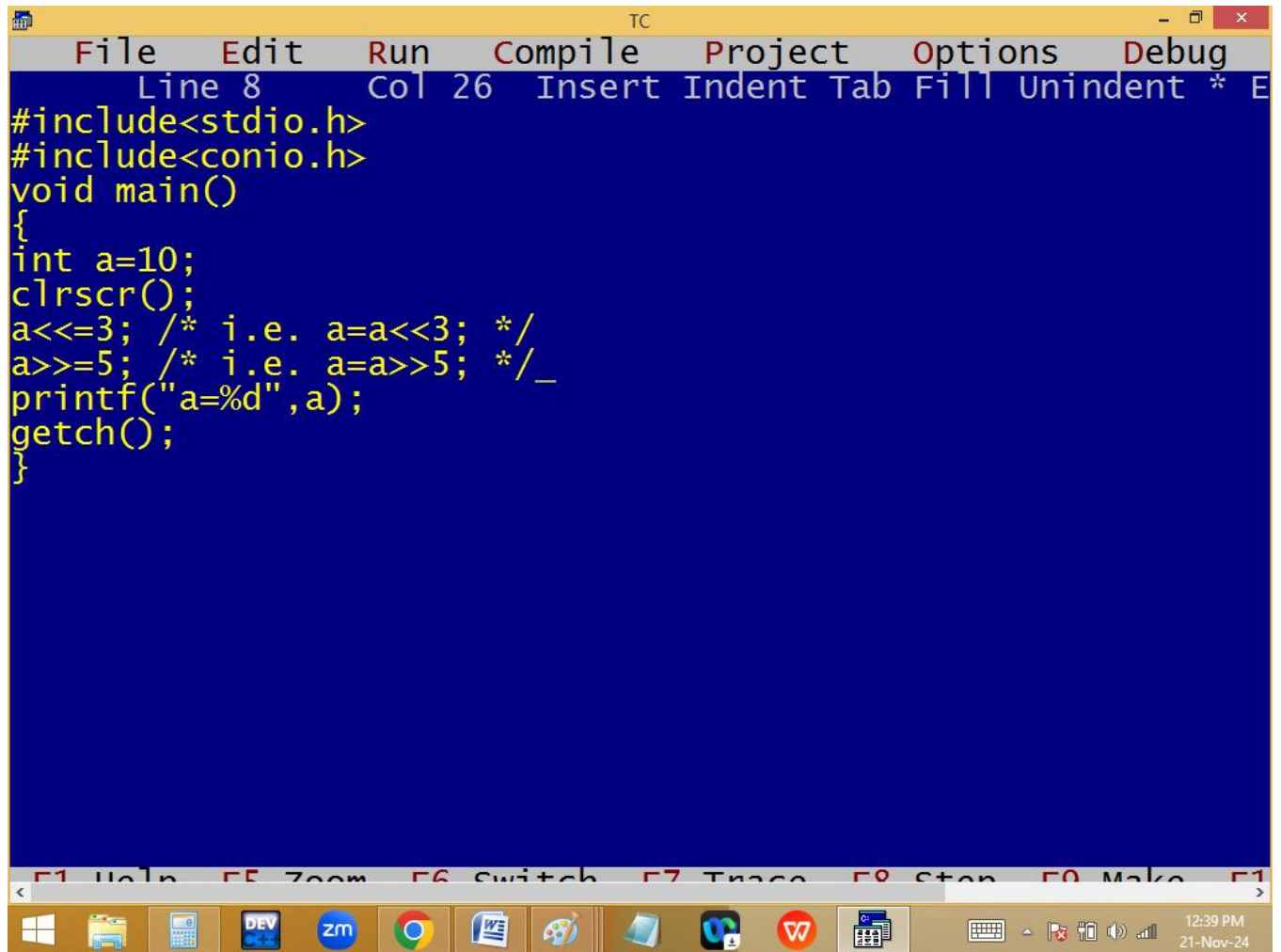


```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 3 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a=a<<2+1>>2;
printf("a=%d",a);
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

12:37 PM
21-Nov-24

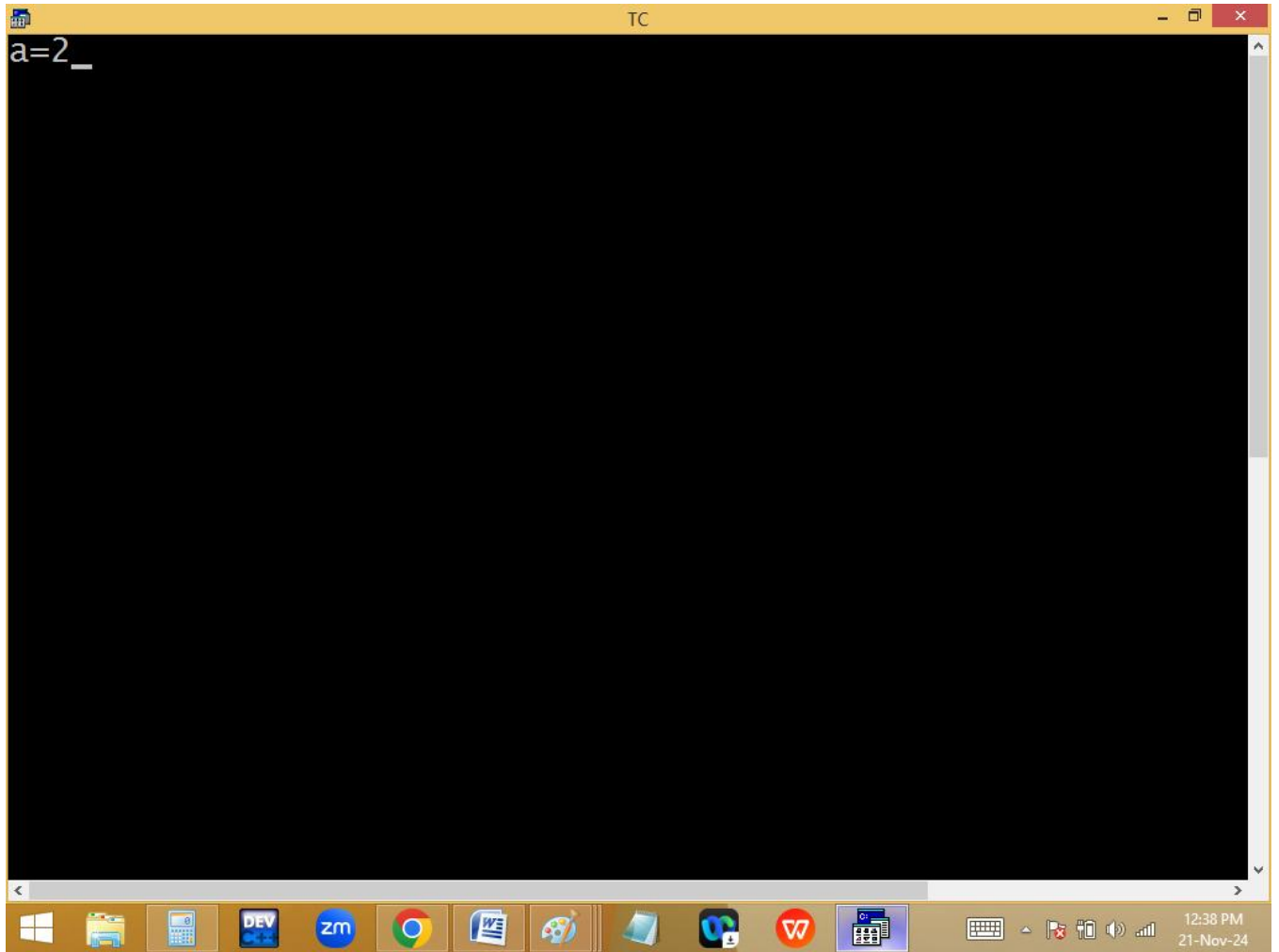


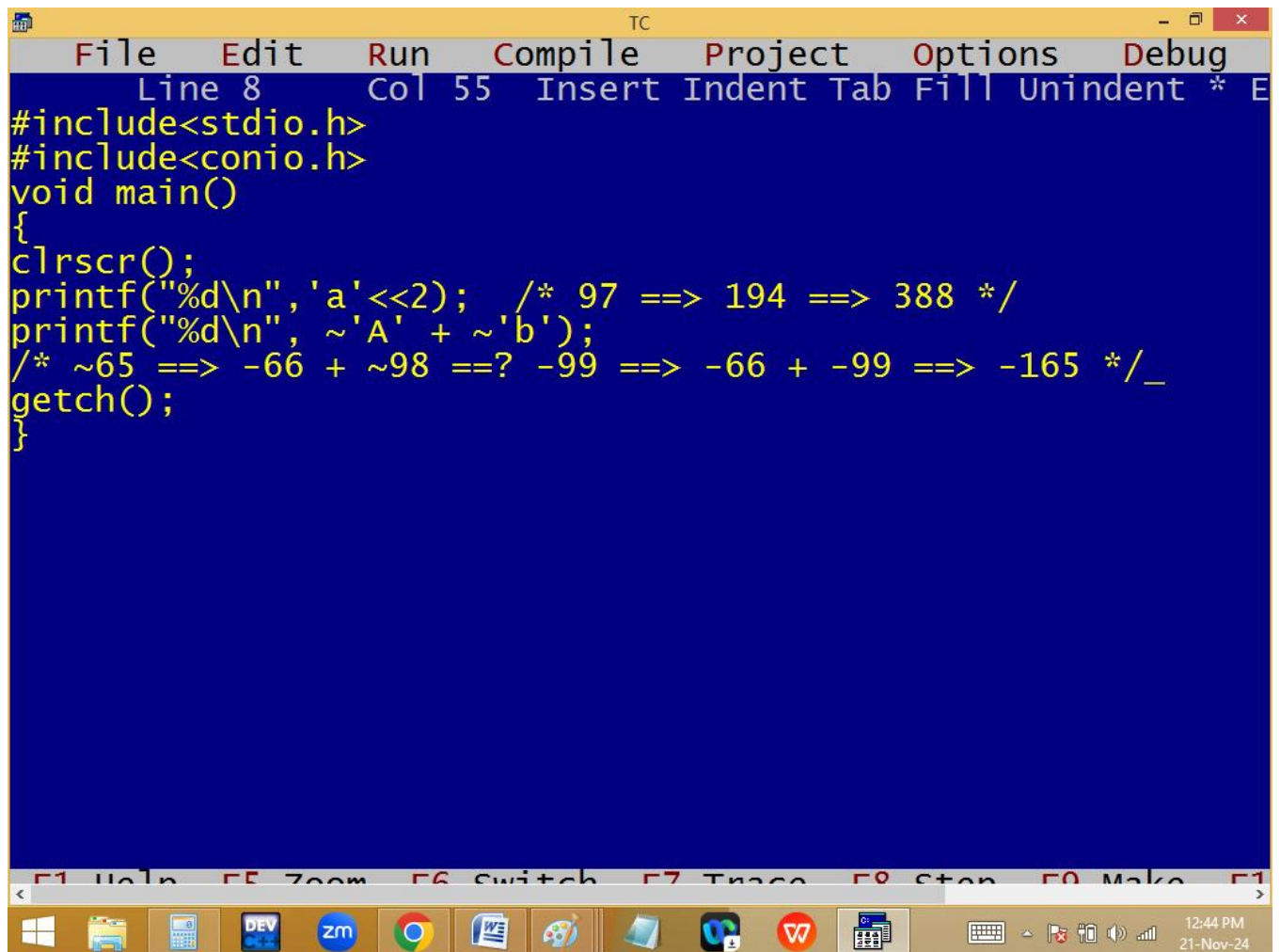


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 8", "Col 26", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10;
clrscr();
a<<=3; /* i.e. a=a<<3; */
a>>=5; /* i.e. a=a>>5; */
printf("a=%d",a);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the bottom, showing icons for the Start menu, File Explorer, DEV, Zoom (zm), Google Chrome, Word, and other applications. The system clock in the bottom right corner shows "12:39 PM" and "21-Nov-24".





```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 55 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("%d\n", 'a'<<2); /* 97 ==> 194 ==> 388 */
printf("%d\n", ~'A' + ~'b');
/* ~65 ==> -66 + ~98 ==? -99 ==> -66 + -99 ==> -165 */_
getch();
}
```

F1 Help F5 Zoom F6 Switch F7 Trace F8 Step F9 Make F10

12:44 PM 21-Nov-24

